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Streamlit-Based Multi-Disease Prediction System for Early Detection of Kidney Disease, Breast Cancer, and Heart Disease

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***Abstract*—The early detection and rapid identification of life-threatening diseases play crucial roles in saving lives. However, disparities in healthcare infrastructure and the unequal distribution of medical resources often hinder timely diagnosis, resulting in adverse outcomes. This study introduces a Multi-Disease Prediction System, a user-friendly web application developed with Streamlit, capable of predicting kidney disease, breast cancer, and heart disease. Unlike traditional AI models that focus on predicting a single disease, our objective is to develop a unified platform using Streamlit and Machine learning to predict multiple diseases simultaneously. The system harnesses advanced machine learning techniques, utilizing algorithms such as the Random Forest, Gradient Boosting Classifier, XGBoost, K-Nearest Neighbor, and Decision Tree classifier. For each disease, the optimal algorithm was selected based on its performance, yielding predictive accuracies of 99% for kidney disease (Random Forest Classifier), 98% for breast cancer (XGBoost), and 95% for heart disease (Random Forest Classifier). Users can input health indicators and symptoms and receive personalized risk insights. By offering an intuitive and interactive platform for data-driven screening, this application seeks to enhance early detection and encourage proactive health management in diverse populations. These models are designed to continuously improve through advancements in machine learning, ensuring reliable and impactful health predictions for broader accessibility and preventive care. Additionally, future improvements, such as integrating more diseases into the system, would further enhance its capabilities and expand its impact on early disease detection.**

Keywords—Disease prediction, Machine learning, Healthcare access, Heart Disease, Breast Cancer, Kidney Disease, Healthcare Prediction.

I. INTRODUCTION

Global health faces substantial challenges from long-term illnesses, such as kidney disease, breast cancer, and heart disease. These chronic conditions significantly contribute to the worldwide burden of death and illness. Early detection and intervention are critical to effectively managing these conditions, as timely diagnosis can improve patient outcomes and quality of life while reducing the long-term burden on healthcare systems. However, conventional diagnostic methods often require specialized facilities and in-person consultations, which may not be accessible to all individuals, particularly in underserved regions. This study leveraged machine learning (ML) to build predictive models for kidney disease, breast cancer, and heart disease, with the aim of developing a unified and accessible platform for risk assessment and early disease detection. Using the Streamlit Python framework, the project introduced an interactive web interface that provides users with an intuitive and practical tool for personalized health management.

A variety of advanced machine learning techniques, including Random Forest, Decision Tree Classifier, K-Nearest Neighbor method, Gradient Boosting Classifier, and XGboost, are integrated into the suggested system to analyze patient data and produce predictions. The system ensures high reliability of its predictions by selecting the most accurate model for each disease. These results underscore the potential of the system to make precise and reliable disease predictions across multiple conditions, thereby providing users with valuable insights into their health. The user interface, built on the StreamLit platform, makes disease prediction accessible to a broad audience, including individuals without technical expertise.

Furthermore, this project tackles healthcare access challenges by providing a remote, data-driven diagnostic tool accessible from any location, reducing dependence on traditional healthcare infrastructure. By enhancing accessibility, especially in resource-limited regions, this system brings disease prediction closer to underserved populations. Through advanced machine learning techniques, the project exemplifies how technology can transform preventive healthcare, helping bridge gaps in healthcare access. It not only supports early diagnosis but also promotes health awareness, encouraging proactive health management. Overall, this innovative digital health solution showcases the potential of machine learning to improve patient outcomes and redefine healthcare accessibility.

II. LITERATURE SURVEY

This section of our work addresses the analysis of previous disease prediction models associated with the presented study. The prediction of heart disease uses two main machine learning methods including logistic regression and SVM for processing clinical diagnostic data and patient symptoms and medical history. Research studies exist about early disease detection and they evaluate multiple models to enhance cardiovascular disease prediction accuracy. Research demonstrates how machine learning can demonstrate improvements in both diagnostic precision and healthcare results for patients. [1]

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Ahire et al. [2] performed study using 700 cases obtained from UCI Machine Learning Repository. A dataset containing 700 cases was obtained from the UCI Machine Learning Repository to execute the study. Classification techniques like SVM, Random Forest, and This study predicts heart disease through the execution of Artificial Neural Network (ANN). The analysis from their study indicated Random Forest demonstrated the minimum accuracy level of 80.0%. Research findings demonstrated that Random Forest achieved an accuracy of 80.0% but SVM surpassed all other models with 84.0% [2].

Agarwal [3] demonstrated an improved method to detect, predict and evaluate breast cancer treatment outcomes through SVM that reached 97.13% accuracy in the algorithm review with Decision Tree, Random Forest, SVM, and Logistic Regression.

The research proposes a single framework to forecast diseases through machine learning classification models including Support Vector Machine (SVM) as well as Logistic Regression alongside K-Nearest Neighbor (KNN). The technology streamlines diagnosis operations and reduces medical costs through its single platform interface which provides forecasts for diabetes cancer and heart disease. The research evaluates prediction accuracy between SVM along with ANN and KNN as well as Decision Trees in diagnosing cardiac disease. The multiple disease prediction system holds importance because it integrates a single platform for disease assessment while enhancing access to medical solutions. PCA functions together with data mining as well as machine learning algorithms to deliver effective analysis [4].

A machine learning-powered system for diagnosing multiple diseases simultaneously is the main subject of "Multiple Disease Prediction System." The approach addresses current healthcare systems' individual-disease orientation since they produce delays or ineffective diagnostic outcomes. The prediction accuracy reached 88% for Random Forest and 89% for XGBoost when detecting diabetes. The heart disease prediction results from Random Forest and KNN demonstrated 85% accuracy and 77% accuracy. The accuracy rates for Random Forest and XGBoost in diagnosing liver illnesses amounted to 73% and 68%. Users benefit from this technology because it provides precise immediate disease predictions for various diseases through a single website and eliminates the need for multiple platforms. The implementation relies on machine learning algorithms Bayesian Decision Tree among XGBoost and KNN [5].

The research develops MDPS which operates through the web to forecast seven common health conditions with deep learning algorithms and machine learning techniques. The system provides accurate predictions through structured and image-based data analysis done with Random Forest and CNN models working together. The platform improves healthcare access through its functions that enable real-time distance monitoring of disease risks while promoting disease awareness and early treatment. Flask and TensorFlow and Keras work together in this system to deliver diagnostic tools that give dependable and user-friendly capabilities for promoting proactive health care [6].

The "Multiple-Disease Prediction System" utilizes ML algorithms to make early detections of Parkinson's disease and heart disease and diabetes and breast cancer. When employed with PCA methodology SVM and Random Forest produce better model performance than other methods. The web application presents a user-friendly interface while performing constraint checks to fill the gaps in multi-disease prediction. [7]

Current studies in ML-based disease prediction identify major weaknesses in their operation which include limited disease scope, restricted accessibility and absence of model performance analysis with no access to real-time remote diagnosis. A user-friendly Streamlit-based system for kidney disease and heart disease as well as breast cancer prediction has been developed to overcome the identified gaps. Predictive reliability together with healthcare accessibility increases through choosing accurate models and ensuring system availability which benefits underserved populations.

III. PROPOSED SYSTEM

Fig 1 discusses the use of various machine learning algorithms, including, Decision Tree Classifier, Random Forest, K-nearest neighbor algorithm, Gradient Boosting Classifier and XGboost to predict multiple diseases. The aim was to bridge the gap between patients and doctors, allowing each to pursue their goals. The accuracy of each algorithm was evaluated and compared to determine the most effective for prediction. Multiple datasets are combined to enhance the accuracy of the predicted results. A user-friendly web application was developed that enables individuals to easily predict diseases by inputting specific attribute values.

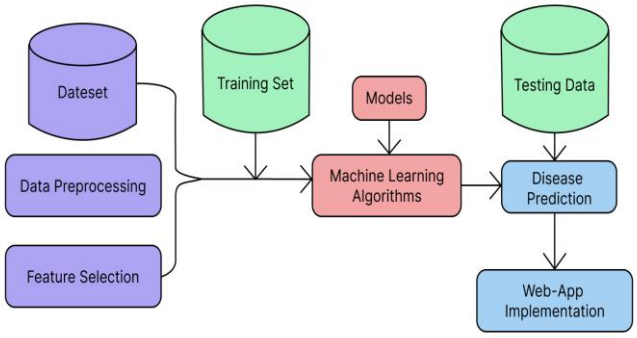


Fig. 1. Workflow of Proposed work

A. Datasets Used

Table 1 displays the datasets that we collect for each of our models.

TABLE I. DESCRIPTION OF EACH DATASET FOR OUR PREDICTION SYSTEMS

Disease prediction system	Dataset Name	Names of medical Parameters	Number of Instance
Heart disease prediction	"Heart Disease Dataset" by Kaggle	14 (including target variable)	606 instances
Kidney disease prediction	"Kidney Disease Dataset" by Kaggle	25 (including target variable)	400 instances
Breast Cancer Predication	"Breast Cancer Wisconsin Dataset" by UCI	32 (including target variable)	569 instances

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B. *Data Preprocessing*

Data preparation is the process of converting raw data into an understandable format by reduction, transformation, and cleaning procedures. Necessary missing data points received imputation through mean and mode substitution for variables that were continuous or categorical respectively. The LabelEncoder() function from sklearn.preprocessing encoded all categorical features for this application. The analysis included an Exploratory Data Analysis (EDA) to show data distributions as well as discover patterns of missing information and locate outliers and measure correlations between features. The analysis used correlation matrices to detect features which showed excessive correlation therefore they were eliminated. SMOTE enabled us to handle class imbalance by generating synthetic minority examples which provided balanced training conditions.

C. *Splitting Train-Test*

The dataset went through preprocessing then split into training (80%) along with testing (20%) sections through train_test_split() with stratification setting the value as Y for keeping class distributions balanced. A model validation through k-fold cross-validation (k=5) was conducted to prevent overfitting and enhance generalization.

D. *Model Training*

After splitting our dataset, we first use the training data to train our model using a variety of machine learning techniques and classifiers. For any prediction system in our situation, we use the Random Forest, Decision Tree Classifier, k-nearest neighbor algorithm, Gradient Boosting Classifier and XGboost techniques to train the model. Hyperparameter tuning was performed using GridSearchCV optimizing parameters like learning rate, depth, and estimators to obtain better accuracy levels.

E. *Model Evaluation and Optimal Model Selection*

Model performance is evaluated using parameters such as accuracy, precision, recall, and F1score to ensure accurate a nd robust results. The best model is chosen after its accuracy is compared to other models. We proceed to Model Prediction, where the model is tested for custom inputs, using the best machine learning method and the testing data.

F. *Model Deployment*

After saving each model, we use Pickle to load them into our web app's database. Finally, we use "streamlit" to develop the web app. and "streamlit-options-menu", and then install our model. We use constraint checks, such as drop-down columns and error pop-ups, to guarantee that users provide consistent data to our models.

IV. RESULTS AND DISCUSSION

A. *Insights from Exploratory Data Analysis*

After conducting Exploratory Data Analysis (EDA), missing values were handled, and new features that might be more useful for the model were created. Categorical variables were encoded using label encoding. Additionally, to address multicollinearity, highly correlated features were removed to ensure better model performance.

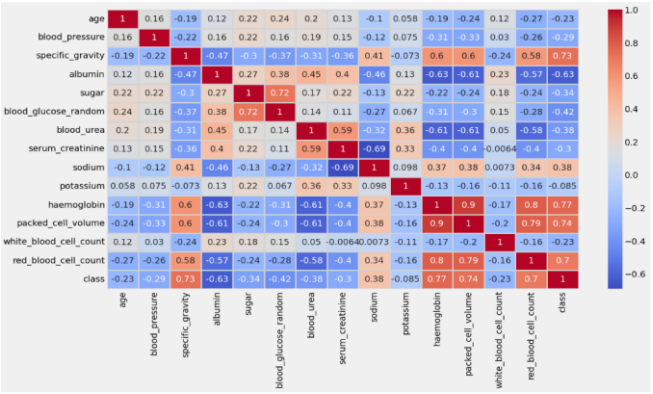


Fig. 2. Heatmap for kidney disease prediction

Key Insights from the correlation Heatmap for Kidney Prediction :

- Strong Relationships: A strong positive correlation exists between haemoglobin and packed_cell_volume, as expected due to their physiological relationship.
- Kidney Function: blood_urea and serum_creatinine show a strong negative correlation, indicating their inverse relationship in kidney function. Albumin levels also negatively correlate with these markers, suggesting its role in kidney health.
- Blood Glucose: Elevated blood_glucose_random levels are associated with higher blood_urea and serum_creatinine.
- Class Prediction: specific_gravity, haemoglobin, and packed_cell_volume strongly correlate with the class, suggesting their potential importance for classification.

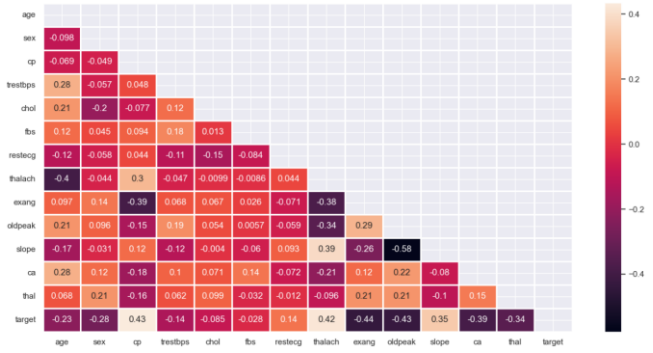


Fig. 3. Heatmap for heart disease prediction

Key Insights from the correlation heatmap for heart disease prediction :

- Strong Predictors: Chest pain type, maximum heart rate, and ST depression are strongly correlated with the presence of heart disease.
- Age and Sex: Age and sex (likely male) are associated with a slightly increased risk of heart disease.
- Blood Pressure and Cholesterol: Resting blood pressure and cholesterol levels have weaker correlations with the target variable.

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- Predictor Relationships: Chest pain type is negatively correlated with ST depression, and maximum heart rate is negatively correlated with exercise-induced angina.

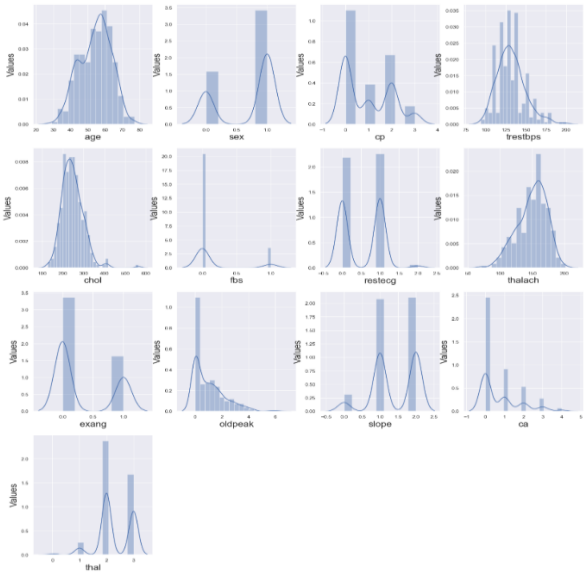


Fig. 4. Histograms for heart disease prediction

Key Insights from the histograms for heart disease :

- Age & Sex: The majority of patients are middle-aged or older, with a slight male predominance.
- Chest Pain: Typical and atypical angina are the most common chest pain types.
- Blood Pressure & Cholesterol: Both resting blood pressure and cholesterol levels tend to have normal distributions with slight right skews.
- Fasting Blood Sugar: Fasting blood sugar levels in the majority of patients are less than 120 mg/dl.
- ECG Results: The majority of patients have normal resting ECG results.
- Exercise-Related Factors: Most patients do not experience exercise-induced angina or significant ST depression during exercise.
- Vessel Coloration & Thallium Stress: A majority of patients have 0 or 1 major vessels colored by fluoroscopy and have normal thallium stress results.

Key Insights from the correlation heatmap for breast cancer prediction :

- Strong Predictors: Several features are strongly associated with malignancy, including radius, perimeter, area, concavity, concave points, and texture. These features provide valuable information for predicting the likelihood of a tumor being malignant.
- Feature Relationships: Strong correlations exist between features like radius, perimeter, and area, indicating their interdependencies.
- Mean vs. Worst: High correlations between mean and worst values of features suggest that tumors with larger or more irregular characteristics in the mean measurements tend to have more severe or advanced characteristics in the worst measurements.

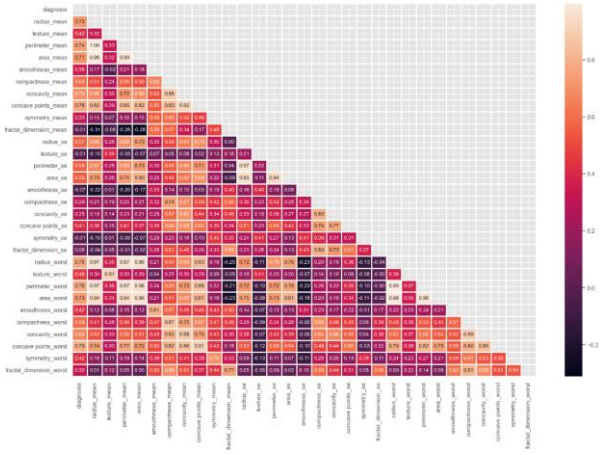


Fig. 5. Heatmap for breast cancer prediction

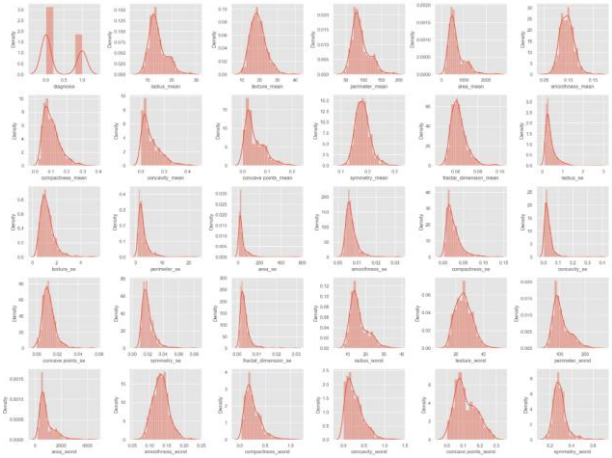


Fig. 6. Histograms for breast cancer disease prediction

Key Insights from the histograms for breast cancer:

- Distribution Shapes: Many features exhibit right-skewed distributions, indicating potential outliers. Some features have approximately normal distributions.
- Outliers: The right-skewed distributions suggest the presence of potential outliers that require careful handling.
- Feature Relationships: While not directly evident from histograms, they can hint at potential relationships between variables.

B. Algorithm Accuracies: Disease Comparison

1) XGBoost: Table 1 shows the highest accuracy (98.24) for breast cancer.

TABLE II. RESULTS OBTAINED USING XGBOOST

Disease Prediction	Accuracy
Heart Disease	94.50
Chronic Kidney Disease	95.83
Breast Cancer	98.24

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2) Decision Tree Classifier: Table 2 shows the highest accuracy (97.50) for kidney disease.

TABLE III. RESULTS OBTAINED USING DECISION TREE CLASSIFIER

Disease Prediction	Accuracy
Heart Disease	92.30
Kidney Disease	97.50
Breast Cancer	92.98

3) Gradient Boosting Classifier: Table 1 shows the highest accuracy (97.50) for breast cancer.

TABLE IV. RESULTS OBTAINED USING GRADIENT BOOSTING CLASSIFIER

Disease Prediction	Accuracy
Heart Disease	92.85
Chronic Kidney Disease	97.50
Breast Cancer	96.49

4) K-nearest neighbor algorithm: Table 1 shows the highest accuracy (98.24) for breast cancer.

TABLE V. RESULTS OBTAINED USING KNN

Disease Prediction	Accuracy
Heart Disease	81.86
Chronic Kidney Disease	63.33
Breast Cancer	91.81

*5) Random Forest Classifier:*Table 1 shows the highest accuracy (99.16) for kidney disease.

TABLE VI. RESULTS OBTAINED USING RANDOM FOREST CLASSIFIER

Disease Prediction	Accuracy
Heart Disease	95.60
Chronic Kidney Disease	99.16
Breast Cancer	97.07

High accuracy rates emerge from Random Forest and XGBoost but these two algorithms surpass K-Nearest Neighbors (KNN) in terms of required computational capacity. The extended complexity of ensemble methods requires longer processing time and increased resource consumption that needs examination before deploying them in real-time systems. Decision Tree and KNN achieve lower performance levels yet they simplify interpretation and need less computing resources which makes them appropriate for lightweight applications.

C. Optimal model selection for each prediction system

TABLE VII. OPTIMAL MODEL SELECTION

Forecasting system	ML Model Used	Test Accuracy
Heart Disease	RFC	95.60
Kidney Disease	RFC	99.16
Breast Cancer	XGB	98.24

The table represents the best-performing models selected after training and evaluating multiple models (Random Forest, Decision Tree Classifier, k-nearest neighbor algorithm, Gradient Boosting Classifier, and XGBoost) on each disease dataset.

The models demonstrate superior accuracy when working with the provided datasets yet their practical implementation could face challenges because of bias issues in the data registry. Future research should utilize expanded training datasets comprising diverse patient populations because it will tackle biases and enhance model performance across different populations which will boost practical generalization abilities.

D. Web-App Stimulation Model Deployment

Our multi-disease forecasting system has its web-based interface built through "streamlit" Python library deployment for optimized user experience. To access multiple web pages, we incorporated the "streamlit options-menu" library into our program. Each webpage includes specific prediction algorithm titles while the input areas function as attribute names. The system indicates the "prediction result" button for users to submit their test requests after completing required inputs. In specific predictive models the input field type ambiguity between integer, floating-point or string values gets validated. These systems present pre-text guidance to users immediately before the input fields and dropdown menus which contain required entry areas.

V. CONCUSION AND FUTURE SCOPE

The proposed Multi-Disease Prediction System shows computer learning methods that improve healthcare accessibility by identifying diseases early. The system leverages advanced algorithms along with a user-friendly web application that Streamlit created to deliver immediate accurate evaluations of kidney disease and breast cancer and heart disease. The platform completes missing capabilities found in typical diagnostic approaches through which users specifically in underserved regions can obtain meaningful health data. The integration of several diseases into one framework improves its practical value while increasing operational efficiency thus providing a major advancement for data-driven accessible healthcare solutions. The system has the potential for transformation in preventive health management due to ongoing development activities along with user feedback implementation.

Future Advancements:

- In order to improve generalizability and lessen biases, future iterations of the system will use larger, more diverse datasets from several sources.
- For more reliable predictions, use deep learning methods like Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) to evaluate complicated data, such as time-series data and medical imaging.
- To further improve accessibility and allow for on-the-go health assessments, develop a mobile application version of the system.
- To ensure inclusivity and serve a varied user base worldwide, include support for many languages.

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